Artificial Neural Networks and Deep Architectures, DD2437

Short report on lab assignment 3: Hopfield networks

Livia Cardaccia, Giovanni Prete

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1 Main objectives and scope of the assignment

The main goals of the assignment were:

- to implement a discrete Hopfield network using Hebbian learning and study its attractor dynamics;
- to investigate synchronous and asynchronous recall, pattern completion, and the role of the energy function;
- to quantify robustness to distortion and investigate how storage capacity depends on the number and statistical properties of stored patterns;
- to study how sparse representations and an activity-dependent learning rule affect associative memory.

The experiments considered bipolar neurons for the standard Hopfield network and binary neurons for the sparse-pattern experiments. Most stochastic experiments used a fixed random seed of 42 for reproducibility. The image experiments used 11 patterns of $32 \times 32 = 1024$ neurons.

2 Methods

All experiments were implemented from scratch in Python using NumPy, with Matplotlib for visualisation. For bipolar patterns, Hebbian weights were computed as

$$W = \sum_{\mu} x^{\mu}(x^{\mu})^{T},$$

and recall used either synchronous updates, $x_i \leftarrow \text{sign}(\sum_j w_{ij}x_j)$, or asynchronous updates of one randomly selected neuron at a time.

Performance was mainly measured by exact pattern recovery, Hamming distance, number of stable memories, and basin size. For the distortion experiments, 100 independently corrupted patterns were tested at each noise level. Sparse binary memories used the centred learning rule based on the mean activity ρ and a threshold θ.

3 Results and discussion: Mechanics & Dynamics

3.1 Convergence and attractors

The three eight-dimensional training patterns x_1, x_2, x_3 were all stable fixed points after Hebbian learning. The network also recovered the supplied distorted inputs, demonstrating content-addressable recall from incomplete information.

However, the stored patterns were not the only stable states. Exhaustive enumeration of all $2^8 = 256$ possible states revealed **14 fixed-point attractors**. Their basin sizes varied considerably, from only 1 state for the smallest basin to 57 for the largest one. Thus, Hebbian learning creates additional spurious minima rather than an energy landscape containing only the intended memories.

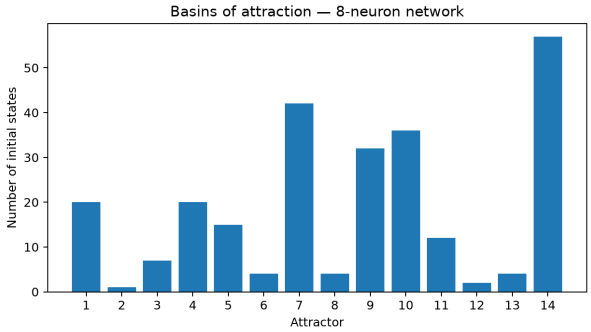


Figure 1: Number of initial states belonging to each of the 14 fixed-point attractors in the eight-neuron network. The strongly unequal basin sizes show that some memories or spurious states are considerably easier to reach than others.

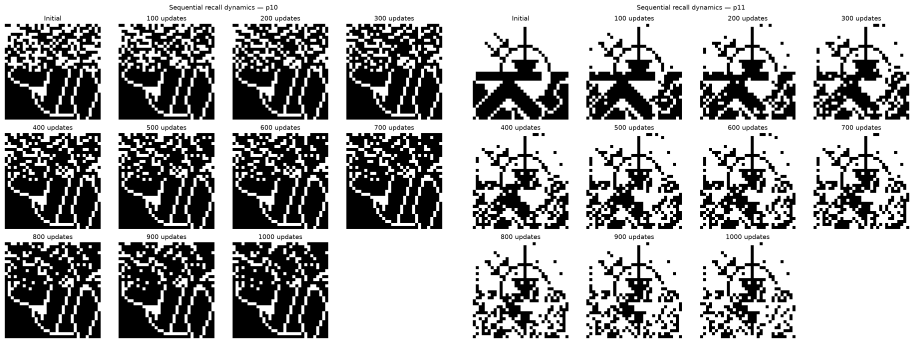
Increasing the Hamming distance from a stored pattern eventually moved the initial state outside its basin of attraction. In particular, once more than approximately half of the bits are incorrect, recall is no longer expected to return the original memory. This illustrates that associative recall depends not only on whether a pattern is stored, but also on the geometry and size of its basin.

3.2 Sequential Update

The larger experiment used p_1, p_2, p_3 as memories in a 1024-neuron network. All three were stable. We then applied asynchronous recall to p_{10}, a degraded version of p_1, and p_{11}, a mixture of stored patterns. A single randomly selected neuron was updated at each iteration.

After 1000 updates, p_{10} was closest to p_1 with Hamming distance 84, while p_{11} was closest to p_3 with distance 181. Neither was exactly equal to a stored pattern at that point. This is not surprising since 1000 asynchronous updates correspond to fewer than one update per neuron on average in a 1024-unit

network.



(a) Sequential recall of p_{10}.

(b) Sequential recall of p_{11}.

Figure 2: Evolution under asynchronous updates. Changes are gradual compared with the practically instantaneous matrix-based synchronous update.

The experiment makes the distinction between the two update schemes clear: synchronous recall changes all 1024 states at once, whereas sequential dynamics reveal the trajectory towards an attractor and are therefore substantially slower when iterations are counted as individual neuron updates.

3.3 Energy

For the small network, the stored patterns had energies

$$E(x_1) = E(x_2) = -34, \qquad E(x_3) = -36,$$

whereas the distorted inputs had higher energies $-20, -18$, and -12, respectively. The intended memories therefore occupy lower-energy configurations than their corrupted versions.

The same behaviour appeared during sequential image recall. For p_{10} the energy decreased from -212982 to -477110, and for p_{11} from -88832 to -350902. No increases were observed in either trajectory; most individual updates left the energy unchanged, while 140 and 237 updates respectively reduced it.

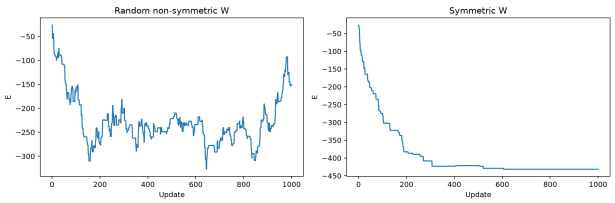


Figure 3: Energy evolution for a random non-symmetric weight matrix and its symmetrised counterpart. Symmetry makes the dynamics substantially more consistent with descent in an energy landscape.

With normally distributed non-symmetric random weights, 152 energy increases were observed in 1000 updates, showing that the quantity is not a Lyapunov function for arbitrary connections. Symmetrisation reduced this to only 7 increases. The residual increases are attributable to the experimental implementation retaining random diagonal self-connections while also including them in the neuron update. In the standard Hopfield convergence argument, setting $w_{ii} = 0$ removes this complication. The result nevertheless clearly demonstrates the qualitative effect of symmetry.

4 Results and discussion: Robustness & Capacity

4.1 Distortion Resistance

Robustness was evaluated by randomly flipping between 0% and 100% of the 1024 neurons of each of p_1, p_2, p_3. Each noise level was evaluated over 100 trials.

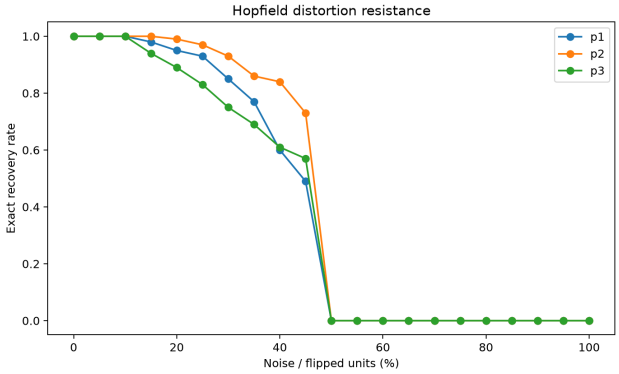


Figure 4: Exact recovery probability versus fraction of randomly flipped neurons for the three stored images.

As shown in Fig. 4, recall was perfect up to 10% noise for all three patterns and remained reliable under considerably stronger distortion. At 30% noise, exact recovery rates were 0.85, 0.93 and 0.75 for p_1, p_2, p_3, respectively. At 40% they were 0.60, 0.84 and 0.61. The transition at 50% was abrupt: exact recovery became zero for every pattern.

This behaviour is consistent with attractor-based recall. Below 50% corruption the noisy input generally remains closer to the original memory than to its inverse or competing states. Around 50%, that information is lost and the input can fall into a different basin. The different recovery curves also show that robustness is pattern-dependent because the memories do not have identical basins of attraction.

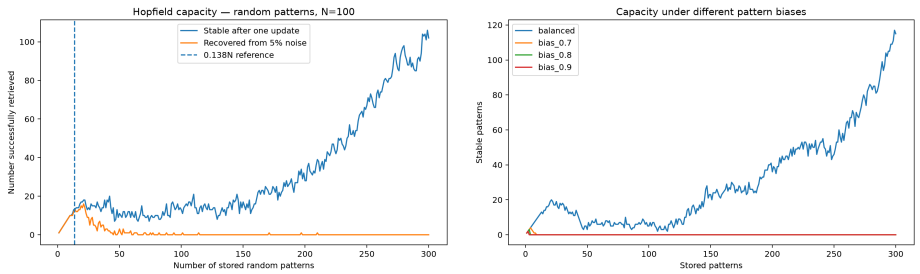
One synchronous update and iterative synchronous recall produced almost identical results. For example, at 40% noise for p_2 the exact recovery rates were 0.81

and 0.86, respectively, while for most other conditions the difference was negligible. Thus, with only three strongly stored patterns, most correction occurred during the first update.

4.2 Capacity

Capacity depended strongly on both the number and statistics of the memories. For the picture data, all of p_1, p_2, p_3 were stable, but after adding p_4 none of the first four patterns remained exact fixed points. This sharp deterioration indicates strong correlations among the structured image patterns.

Random bipolar patterns behaved substantially better. For $N = 100$, all memories were stable up to 10 stored patterns. Performance then degraded gradually: at 20 memories, 17/20 were stable and 14/20 could be recovered from 5% noise; at 30 memories, these values dropped to 15/30 and 5/30. No noisy retrievals remained by approximately 45 stored patterns.



(a) Random bipolar memories.

(b) Effect of pattern bias.

Figure 5: Storage capacity depends not only on the number of memories but also strongly on their statistical distribution.

Self-connections gave an apparently higher stability capacity. Without them, the number of stable patterns fell to zero at 33 stored patterns, whereas retaining them allowed many states to remain stable even at much larger loads. This apparent advantage is misleading for associative recall: a positive w_{ii} encourages a neuron to retain its current value, which can stabilise incorrect or spurious states rather than correct noise.

Pattern bias had an even stronger effect. With 300 random balanced patterns, 115 were still fixed points in the final overloaded network, whereas distributions with probabilities 0.7, 0.8 or 0.9 of +1 produced no stable stored patterns at this load. The result also helps explain the poor picture-pattern capacity: correlated or statistically unbalanced memories generate substantially more interference than independent balanced patterns.

4.3 Sparse Patterns

To compensate for the poor capacity of biased representations, binary sparse patterns were trained using centred activities $(x_i - \rho)$ and a threshold θ. The threshold sweep showed that the useful range of θ depends strongly on the activity level. For $\rho = 0.10$, the subsequent capacity experiment selected $\theta = 0.15$, while the very sparse $\rho = 0.01$ case selected $\theta = 0.05$.

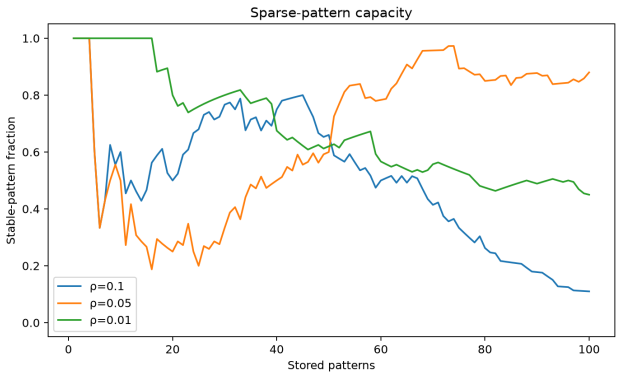


Figure 6: Fraction of stable sparse memories as the number of stored patterns increases for activity levels $\rho = 0.10$, 0.05 and 0.01, using the best threshold obtained from the threshold sweep.

The capacity curves demonstrate that sparsity changes the interference structure substantially. With $\rho = 0.10$, all patterns were stable only up to four memories at $\theta = 0.15$, after which the stable fraction dropped. The sparser representations retained substantial fractions of their memories at much higher loads; for example, the $\rho = 0.01$ experiment still had 45 stable patterns out of 100 at the end of the sweep.

The important observation is therefore that sparse coding must be accompanied by an appropriate learning rule and threshold. Centering the activities prevents the dominant inactive state from producing a large systematic weight bias, while θ controls the activation required during recall.

5 Final remarks

The experiments demonstrated how a Hopfield network can implement associative memory through attractor dynamics rather than explicit input–output mappings. The most interesting result was the connection between three apparently separate properties: energy, robustness and capacity. Hebbian learning creates low-energy memories with finite basins of attraction, allowing substantial noise removal, but increasing the number or correlation of memories reshapes these basins and eventually destroys reliable recall.

The comparison between random, biased and sparse patterns was particularly useful because it showed that storage capacity cannot be described only by the

number of neurons. The statistical structure of the memories is equally important. Likewise, the self-connection experiment illustrated why simply counting fixed points can overestimate useful memory performance.

It remains to be clarified how the empirical capacity curves would change if the average were calculated over many sets of independently generated random models, rather than over a single seeded realization. This would allow confidence intervals to be associated with the capacity transition and would make the comparison with the theoretical asymptotic capacity more rigorous.